

L9: Effective Distance

Can we use geographical distance to predict the time that an epidemic will arrive at a state or country?

Again, we can use epidemic models such as GLEAM to answer such questions.

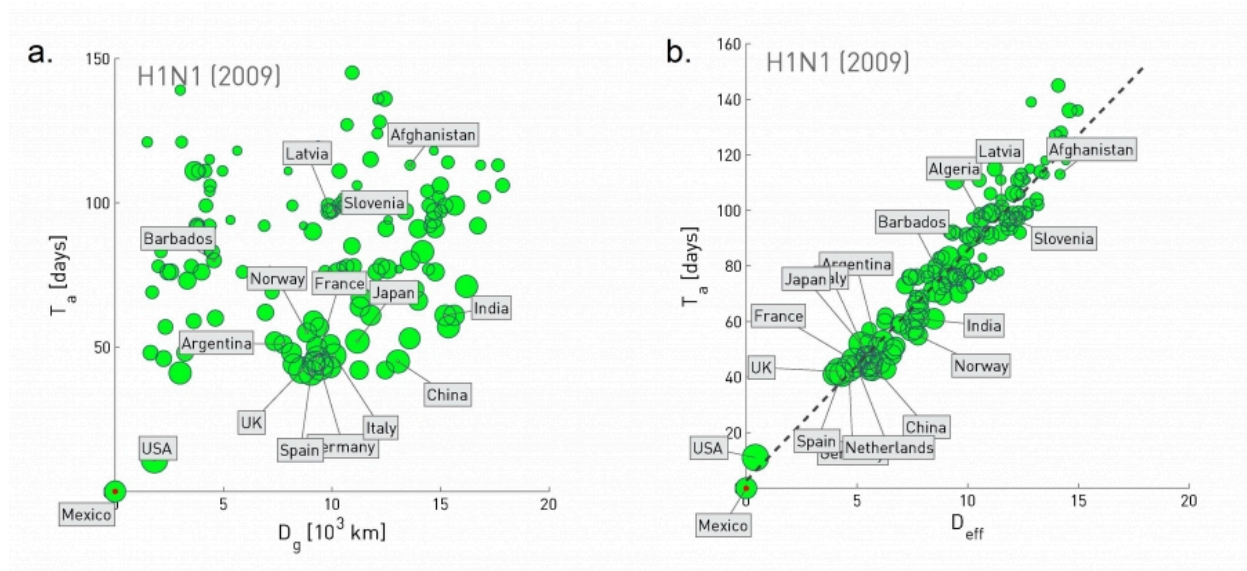


Figure 10.32 from *Network Science* by Albert-László Barabási

The plot at the left shows the geographic distance between Mexico and many other countries at the x-axis, while the y-axis shows the time that the H1N1 pandemic arrived in that country (*defined as the number of days between the first confirmed case in that country and the beginning of the outbreak on March 17, 2009*).

Clearly there is not any strong correlation between the two quantities.

Let us now define a different kind of distance, based on mobility data rather than geography:

Suppose that we have data from airlines, trains, busses, trucks, etc, showing how many travelers go from city i to city j .

The fraction of travelers that leave city i and arrive at city j is denoted by $p_{i,j}$.

The effective distance between the two cities is then calculated as $d_{i,j} = 1 - \ln p_{i,j}$.

The plot at the right replaces geographic distance with “**effective distance**”, and it shows that that the arrival day of this pandemic from Mexico was actually quite predictable based on strictly mobility data.